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Source-Scored Structural Initial Designs for Chance-Constrained Optimization of Ordered Policy Profiles

(Authors' names are not included for peer review)

Abstract. Many policies stored as thousands of ordered settings are finely discretized rules. We study how historical simulations can choose starting policies for a new chance-constrained problem of this kind. Replicated source outcomes rank a public library of 64 profiles by objective and safety; farthest-first selection chooses ten profiles covering a standardized cosine coordinate augmented by these ranks. The design is fixed before target outcomes and followed by an independent deployment verifier. We establish cross-grid stability, finite-library coverage, recovery of separated source ranks, conditional coverage under source-target alignment, and familywise certificate validity. Across the original 480 task–resolution cells, source scoring achieves 47.3% certified true-feasible deployment versus 31.5% for standardized outcome-free maximin. A prospective study on 240 new tasks, three new source families and three public libraries yields 23.3% versus 10.0%. Factorial controls attribute most of the gain to the source-selected first center; rank augmentation adds little. A subsequent fixed-first-center diagnostic gives 53/240 certificates with nine random remainder policies versus 54/240 with structural coverage. A same-source portfolio gives better feasible coverage and penalized loss in the new study, while electricity results do not establish an archive cost advantage. A separate public-battery development replay is matched by a source-free constant-policy search and favors paid enumeration at the source-inclusive budget. The evidence supports source-selected initialization under the declared synthetic task laws; its operational value requires transferable structure and recovery of the archive cost.

Key words: simulation optimization, chance constraints, initial design

Subject classifications: Simulation: initial designs and independent verification; programming, stochastic-chance constrained: ordered policies.

Area of review: Simulation

1. Introduction

Many simulation decisions are rules before they are vectors. A reserve rule changes with forecast stress, an inventory rule changes over a planning horizon, and a resource rule changes across ordered operating states. A computer may store such a rule as hundreds or thousands of numbers, but those numbers are not unrelated knobs. They are samples of one ordered profile. Ignoring that fact can make the problem look much harder than the underlying decision really is.

Chance constraints make the first simulations especially important. If none of them reaches a feasible region, a surrogate has little evidence about where the safety boundary lies. Historical simulations from related systems can help, but only if the relevant policy shapes transfer. A shape that was safe in one system

may be unsafe in another, and the cost of collecting and verifying historical evidence may exceed the cost of the new search itself. The useful question is therefore not simply whether history helps, but what part of history can be reused, under what assumption, and at what cost.

We focus on one concrete use of history: choosing the *initial design* for a new chance-constrained ordered-profile problem. We first construct a public library of 64 constant, monotone, smooth, piecewise, and oscillatory profiles without using outcomes. Replicated source simulations then rank these same profiles by objective and chance margin. The best compromise starts the design, and farthest-first selection adds profiles that cover a common cosine coordinate. The resulting ten policies are fixed and mapped to the target grid before the first target outcome. We call this a *source-scored structural initial design*.

The method keeps three jobs separate. The initial design decides where a new search begins. Any declared optimizer may add target evaluations. Finally, a short ordered list of candidates is fixed and tested with fresh simulations; if none passes, the procedure abstains. Figure 1 shows the decision sequence. We use the ordered profile and history to choose among transparent candidate shapes, and reserve fresh target outcomes for deployment verification.

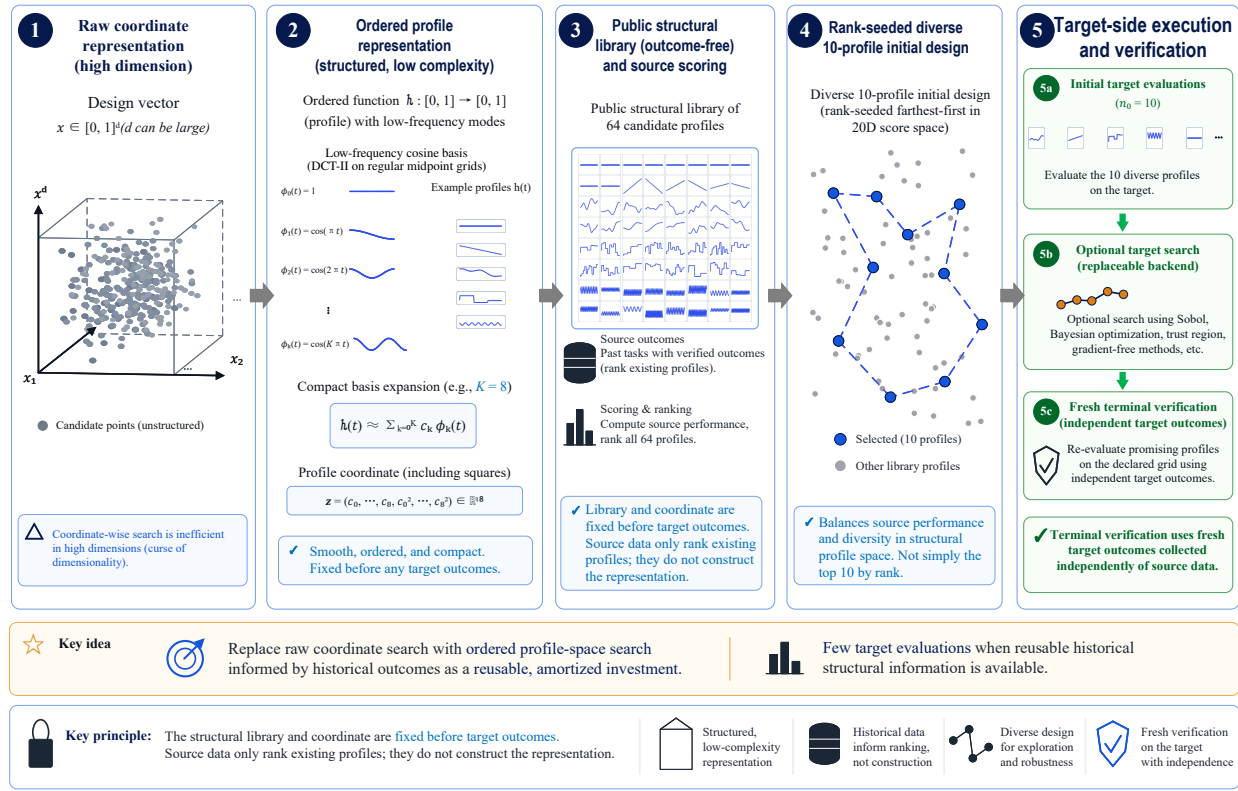


Figure 1 From high-dimensional coordinates to ordered profile-space search. The raw vector is read as samples of an ordered policy and summarized by low-frequency cosine coefficients. The public library and coordinate are fixed without outcomes. Source simulations rank existing profiles, while farthest-first selection prevents the ten-point design from collapsing onto a single source-favored shape. A replaceable optimizer may add target evaluations. Fresh target simulations then verify a fixed shortlist.

The paper makes three contributions.

1. We give a finite, reproducible way to turn source simulations into ten starting policies. The library, coordinate, weights, tie breaks, and target mapping are explicit, and the design uses no target outcome or hidden target geometry.
2. We explain when the small design can work. Grid refinement preserves the profile coordinate; farthest-first controls finite-library cover; finite source samples recover separated ranks, and selection gaps determine whether source noise changes the design. Structural cover reaches a target safe region when transfer error fits inside its safety margin. Fresh tests control false deployment. Lean 4 checks the original finite components, while source-target alignment remains an empirical assumption.
3. We evaluate the resulting design in eight randomized regimes, a factorial first-center/metric intervention, prospective confirmation on new source families and public libraries, a same-source portfolio, equal-cost experiments, postdecision fixed-first-center remainder controls, and a region-held-out electricity study that does not establish a cost advantage. A separate public-battery development diagnostic is matched by

constant-only search and favors paid enumeration at the source-inclusive budget. Source, target-search, and verification calls are reported separately. The evidence measures when source initialization helps under specified task laws and when the archive cost remains unjustified.

Across 160 independently seeded tasks and three grid resolutions, source scoring certifies 47.3% of task-resolution cells. Supplementary comparisons on these same tasks give 31.5% for standardized outcome-free maximin and 37.1% for the source-greedy portfolio. The matched scaling corrects a confound in the original outcome-free comparison, and the portfolio tests whether geometric selection adds value beyond a simple use of source ranks. The advantage varies by mechanism, with reversals in adverse regimes and no established archive cost advantage in Energy. On 240 new tasks the matched generic comparison remains positive, but portfolio superiority is not reproduced. Attribution identifies the source-best first center as the main effective contribution. The fixed-first-center controls do not establish a certification advantage for structural coverage over nine random remainder policies. The 10,000-coordinate result concerns stability under grid refinement of an ordered rule.

2. Related Literature

Simulation optimization and noisy constraints. Simulation optimization studies expensive stochastic systems for which objectives or constraints are available only through simulation (Fu 2002, Amaran et al. 2016). Stochastic kriging explicitly separates response uncertainty from simulation noise (Ankenman et al. 2010), while constrained Bayesian optimization models noisy objective and feasibility observations jointly (Letham et al. 2019). Our question occurs one stage earlier: before fitting a target surrogate, which few ordered profiles should receive the first simulations?

High-dimensional and functional optimization. Trust-region methods and sparse axis-aligned priors can make Bayesian optimization more robust in large ambient spaces (Eriksson et al. 2019, Eriksson and Poloczek 2021, Eriksson and Jankowiak 2021). They do not by themselves encode that coordinates discretize one ordered function. Functional Bayesian optimization instead searches over function-valued decisions (Vien et al. 2018). Functional-data analysis likewise represents curves through common bases and treats discretization as observation of a function rather than unrelated coordinates (Ramsay and Silverman 2005). Our profile coordinate is less flexible but fully finite: it maps irregular or refined grids into a common cosine summary, then selects from a declared structural library. Nominal grid dimension and effective profile complexity are therefore reported separately.

Space-filling initial designs. Maximin and minimax distance criteria are classical model-free devices for covering an experimental region (Johnson et al. 1990), and space-filling designs are standard for deterministic computer experiments (Morris and Mitchell 1995). Gonzalez farthest-first gives a factor-two finite k -center guarantee (Gonzalez 1985). We use that algorithm on a finite profile library, but source ranks augment its selection metric. The standardized DCT maximin control uses the same structural coordinates without source

ranks. It separates source information from the coordinate scaling used by the original unstandardized DCT control.

Learned initializations and portfolios. Learning starting configurations from historical tasks is established work. Wistuba et al. (2015) optimize a meta-loss to learn an initialization sequence, while Feurer et al. (2022) greedily construct a finite portfolio of complementary configurations and evaluate it before continuing Bayesian optimization. Our contribution concerns ordered policies under noisy chance constraints: a common structural coordinate across grids, joint safety and objective source ranks, and independent deployment verification with explicit source costs. To compare selection rules within this setting, we include a source-greedy portfolio using the same archive and ten target evaluations.

Transfer and historical-task priors. Few-shot deep kernels, hierarchical Gaussian-process priors, source ensembles, and meta-learned acquisitions transfer different objects across tasks (Wistuba and Grabocka 2021, Rothfuss et al. 2021a,b, 2023, Wang et al. 2024, Tighineanu et al. 2022, Volpp et al. 2020, Pan et al. 2024). Related work also learns a search space from earlier hyperparameter tasks (Perrone et al. 2019) or weights source surrogates by ranking performance (Feurer et al. 2018). Warm-start Bayesian optimization similarly treats earlier evaluations as reusable information (Poloczek et al. 2016). Our method transfers a finite initial design: the source archive ranks a fixed public library, while the target surrogate and optimizer remain separate. This makes it possible to compare source scoring directly with the same structural library without outcomes and to count the cost of history explicitly. We also compare eight end-to-end transfer pipelines under their own common source, target-budget, and verification protocol.

Selection and independent verification. Chance-constrained ranking-and-selection procedures allocate simulation to identify a feasible best system while controlling selection error (Hong et al. 2015). Post-optimization ranking and selection can also “clean up” candidate solutions generated by a heuristic search (Boesel et al. 2003). We adopt this separation literally. Search produces an ordered shortlist; fresh Bernoulli feasibility replications test its members; and Bonferroni spending composes candidate-wise exact tests. The all-success rule is conservative, so we report power and verification cost in addition to false certificates. This differs from treating a surrogate’s posterior feasibility probability as a deployment certificate.

Positioning. Without restrictions on the objective and feasible set, no finite target-label-free design can dominate all others, a point consistent with no-free-lunch results (Wolpert and Macready 1997). Our claim is therefore about experimental design for related ordered-profile tasks, not a distribution-free optimizer. Source-free structured designs, adverse regimes, an external negative control, and explicit cost accounting show where that claim does and does not hold.

3. Problem Setting and Available Information

3.1. Ordered policy profiles

Let $h : [0, 1] \rightarrow [0, 1]$ denote an ordered policy profile. A task q supplies ordered nodes

$$0 < t_{q,1} < \dots < t_{q,d_q} < 1$$

and an integer implementation scale L_q . The deployed vector is

$$x_{q,i}(h) = \text{round}\{L_q h(t_{q,i})\}, \quad i = 1, \dots, d_q.$$

Thus d_q is a grid resolution, not by itself the intrinsic dimension of the admissible profile class. Regular grids use $t_{q,i} = (i - 1/2)/d_q$; irregular grids are allowed when their ordering is declared. A schema may additionally declare channel order. No claim is made for an unordered vector with unknown coordinate semantics.

One simulator replication at x returns an objective observation $Y_q^f(x)$ and a scalar constraint observation $Y_q^g(x)$. We minimize

$$f_q(x) = \mathbb{E}[Y_q^f(x)]$$

subject to the chance constraint

$$p_q(x) := \mathbb{P}\{Y_q^g(x) \leq \tau_q\} \geq 1 - \alpha_q. \quad (1)$$

The true feasible set is $\mathcal{F}_q = \{x : p_q(x) \geq 1 - \alpha_q\}$. In synthetic experiments, the experimenter reveals truth only after design, search, shortlisting, and verification are complete, and uses it only to score the finished decision.

3.2. Source and target information

Let $q_1, \dots, q_m$ be source tasks and $q_\star$ a target task. Before the target opens, every source task evaluates the same finite normalized profile library $\mathcal{L} = \{h_1, \dots, h_J\}$. For each pair (q, h) , the source archive contains R ordinary replications of (Y_q^f, Y_q^g) . The primary method uses $m = 2$, $J = 64$, and $R = 3$, hence

$$S = mJR = 384 \quad (2)$$

source simulator calls. The library is constructed without source or target outcomes; source data only determine which members are selected.

The source archive is supplied before the target problem opens. In the synthetic stress test its tasks come from the same declared regime as the target; in the external study they obey the stated region-holdout rule. The method ranks profiles within this archive. Choosing or purchasing related historical tasks from a larger repository is a separate upstream decision.

Before target search, the method may use the target's ordered grid, integer bounds, nominal dimension, and, in declared-schema experiments, channel order. It may not use a target outcome, feasibility label, noise parameter, optimum, safe-basin center, or verification sample. The primary source aggregation is domain blind: source tasks receive equal weight. Descriptor-conditioned weighting is evaluated only as a control.

3.3. Decision stages and costs

The initial-design step selects n_0 target policies. A target optimizer may then use a total of $N \geq n_0$ search calls and return an ordered list of at most K_s distinct candidates. This list and its order are fixed before fresh verification simulations begin. The final outcome is either a certified deployment or abstention.

We report the costs

$$C_{\text{all}} = S + N + V, \quad C_{\text{amort}}(M) = \frac{S}{M} + N + V, \quad (3)$$

where V is the realized independent-verification cost and M is the number of target deployments sharing one source archive. “Equal preverification cost” means that $S + N$ is matched before V ; it is not called equal total cost.

3.4. Endpoints

The primary search endpoint is whether the N target evaluations contain a truly feasible policy. The primary deployment endpoint is whether the verifier returns a truly feasible policy. We also report false certificates, verification calls, and a penalized loss that retains failures in the denominator. Conditional regret is reported only when a feasible policy exists and is not used alone to compare methods with different feasible counts. A separate fixed library defines synthetic regret and an unattainable finite-library upper bound; neither affects an implementable decision.

4. Source-Scored Structural Initial Design

The initial design has three steps: construct a public library of policy shapes, summarize each shape in a common coordinate, and use source outcomes to choose a promising but diverse subset. This section specifies each step.

4.1. Outcome-free profile library

All tasks share a deterministic library $\mathcal{L} = \{h_1, \dots, h_{64}\}$ on 128 regular cell midpoints. Its seed is 20261799, obtained by adding 991 to the family seed 20260808. It contains:

- ten constants with levels 0.05, 0.15, $\dots$, 0.95;
- six linear ramps with endpoint pairs (.05, .95), (.95, .05), (.20, .80), (.80, .20), (.35, .65), and (.65, .35);
- sixteen low-frequency cosine profiles, with rank drawn uniformly from $\{2, \dots, 8\}$, coefficient $a_k \sim N(0, .12^2/k)$, and baseline drawn uniformly from $[.25, .75]$;
- sixteen piecewise-constant profiles with 3–12 blocks and levels drawn uniformly from $[.08, .92]$, followed by the declared deterministic smoothing;
- sixteen cosine profiles with frequency drawn uniformly from $\{9, \dots, 40\}$, baseline in $[.30, .70]$, amplitude in $[.08, .25]$, and a uniform phase.

Values are clipped to $[0, 1]$. Duplicate profiles are skipped, and stable profile identifiers follow construction order. This mixture is a structural prior. It is public, generated before any outcome, and is not claimed to span all useful policies.

4.2. Grid-invariant profile coordinate

For ordered nodes $t_1 < \dots < t_d$, let $0 = e_0 < e_1 < \dots < e_d = 1$ be their Voronoi cell edges, where $e_i = (t_i + t_{i+1})/2$ internally. Write $\Delta_{ik} = \sin(\pi k e_i) - \sin(\pi k e_{i-1})$. For the piecewise-constant reconstruction of h , define

$$c_0(h) = \sum_{i=1}^d h(t_i)(e_i - e_{i-1}), \quad (4)$$

$$c_k(h) = \frac{\sqrt{2}}{\pi k} \sum_{i=1}^d h(t_i) \Delta_{ik}, \quad k = 1, \dots, K. \quad (5)$$

These are exact integrals of the cosine basis over the reconstruction cells, not midpoint approximations to the basis. On regular midpoint grids they coincide with DCT-II coefficients up to normalization; on irregular grids we use their exact cell-integral cosine analogue. With $K = 8$ and $\kappa = .25$, the profile coordinate is

$$\phi(h) = (\tilde{c}_0, \dots, \tilde{c}_K, \tilde{c}_0^2, \dots, \tilde{c}_K^2), \quad \tilde{c}_k = \frac{c_k}{1 + \kappa k}. \quad (6)$$

It has dimension 18 and contains no cross-products. Each coordinate is standardized over the 64-profile library before selection. Orthogonality supplies a coordinate system; truncation and the finite profile library impose the complexity restriction. The squared coordinates describe mode strength and do not assume an additive target response. On this library, standardization cancels the positive frequency rescaling in both the coefficient and squared-coordinate columns. Thus κ is redundant in the standardized selection metric; its frozen value specifies the coordinate before standardization.

Here c_0 is the profile's average level, while $c_1, \dots, c_8$ describe its broad upward, downward, and oscillatory shapes. Their squares add a simple measure of mode strength without introducing every pairwise interaction. The cell integrals are what let the same shape receive a comparable description on a 200-, 1000-, or 10,000-point grid.

4.3. Replicated source scores

For source task q , profile j , and replication r , let Y_{qjr}^f and Y_{qjr}^g be the two simulator outputs. Define

$$\hat{f}_{qj} = R^{-1} \sum_{r=1}^R Y_{qjr}^f, \quad (7)$$

$$\hat{m}_{qj} = \bar{Y}_{qj}^g + z_{1-\alpha_q} \max\{s_{qj}^g, 10^{-3}\} - \tau_q, \quad (8)$$

where s_{qj}^g is the sample standard deviation with divisor $R - 1$. Within each task, deterministic average-tie percentile ranks map the 64 objective means and margins to $[0, 1]$, with lower values preferred. Let r_{qj}^f and r_{qj}^g denote those ranks. The primary domain-blind scores are

$$r_j^f = \frac{1}{m} \sum_{q=1}^m r_{qj}^f, \quad r_j^g = \frac{1}{m} \sum_{q=1}^m r_{qj}^g. \quad (9)$$

Whether $\widehat{m}_{qj} \leq 0$ in every source task is logged as a diagnostic but does not filter the library. Descriptor-conditioned source weights are a secondary control and are not part of the primary method.

The margin combines the observed constraint mean and scale in the same units as the threshold: a negative value favors feasibility. Percentile ranks then remove differences in objective and constraint scales across source tasks.

4.4. Rank-seeded farthest-first selection

Let z_j be the standardized version of $\phi(h_j)$. This 18-dimensional coordinate contains structural profile information only. Selection uses the 20-dimensional augmented coordinate

$$\eta_j = (z_j, r_j^g, r_j^f) \quad (10)$$

with Euclidean distance. The first center is

$$j_1 \in \arg \min_j \{.5r_j^g + .5r_j^f\}, \quad (11)$$

breaking ties by r_j^g , then r_j^f , then stable profile identifier. For $\ell = 2, \dots, n_0$,

$$j_\ell \in \arg \max_{j \notin \{j_1, \dots, j_{\ell-1}\}} \min_{s < \ell} \|\eta_j - \eta_{j_s}\|_2, \quad (12)$$

with the lowest library index breaking an exact tie. The primary design uses $n_0 = 10$. Each selected 128-node profile is linearly interpolated onto the declared target nodes, multiplied by L_q , clipped, and deterministically rounded to an integer policy.

Selecting the ten best source ranks alone could return ten nearly identical profiles. Farthest-first avoids that collapse: source performance chooses the starting point and influences distance, while each later point must add new coverage.

The two coordinates play different roles. The structural coordinate z says what a profile looks like. The augmented coordinate η also says how that profile performed on source tasks, so it determines which profiles are selected. The target coverage theorem uses only z : projecting η back to z cannot increase Euclidean distance. We therefore do not pretend that a source percentile rank has a target counterpart.

4.5. Target search and verification

The initial design does not depend on a particular target optimizer. The primary comparison sets $N = n_0 = 10$, so no sequential search can blur the effect of the ten starting policies. Functional SCBO, SAASBO, and native transfer methods are evaluated separately.

Given noisy target records, the neutral protocol freezes up to three distinct shortlist entries in this order: best observed-objective empirical feasible policy, observed safe-interior policy, and minimizer of

$$Y^f(x) + 2[Y^g(x) - \tau]_+, \quad (13)$$

followed by objective-ranked fallbacks if needed. Fresh simulator replications then test shortlist members sequentially. Candidate j with budget v_j is certified only if all v_j constraint replications satisfy the threshold and $(1 - \alpha)^{v_j} \leq \delta/K_s$. The primary budgets are $(80, 80, 80)$, $K_s = 3$, and $\delta = .05$. Failure to certify produces abstention, not a deployment.

4.6. Comparison designs

Standardized DCT maximin uses the same 64 profiles and the same z coordinates as the source-scored atlas, with the coordinate-space medoid as its first center and farthest-first for the remaining nine. It uses no outcomes. The original *generic DCT maximin* implementation instead used unstandardized ϕ throughout; we label its frozen results *unstandardized DCT maximin*. Both coordinate scaling and source information therefore differ in that original comparison.

The *source-greedy portfolio* uses the same source archive to select ten profiles by their complementary source performance. It greedily minimizes the average, over source tasks, of the best selected profile's equal-weight safety and objective rank. The finite rule and ties are specified in the Online Supplement. Random low-frequency, natural three-block, and raw Sobol designs separate other forms of structural prior. The finite-library upper bound is unattainable and is used only to interpret completed experiments.

Thus source data choose among profiles that already exist; they do not learn the library itself. The standardized and portfolio comparisons assess, respectively, the value of source outcomes and the choice of selection rule.

5. Why the Design Can Work: Finite-Sample Guarantees

The method uses only ten target policies, so its structural assumption must do real work. The results below trace that work from representation to deployment. Proposition 1 first shows that a policy has nearly the same coordinate when the implementation grid is refined. Theorem 1 then shows that farthest-first gives a controlled cover of the finite library. Proposition 2 states which source rankings survive finite simulation noise. Theorem 2 turns structural cover into a feasible target policy when a source-supported profile lies

close enough to a sufficiently deep target safe region. Finally, Theorem 3 controls false deployment using fresh target simulations.

This chain also makes the limitation visible. A fine grid does not hurt if it describes the same ordered profile, but no theorem can rescue a target whose safe region is absent from, or badly distorted relative to, the structural library. The results are finite and conditional. The covering guarantee holds for any first center, and source-rank recovery concerns source tasks alone. Neither implies that source scoring outperforms a source-free design or a source-greedy portfolio; that comparison is empirical.

5.1. Profile-grid consistency

Let $c_k(h) = \int_0^1 h(t)b_k(t) dt$, where $b_0(t) = 1$ and $b_k(t) = \sqrt{2} \cos(\pi kt)$ for $k \geq 1$. Let $\widehat{h}_d$ be a reconstruction from d ordered samples.

PROPOSITION 1 (Coefficient stability). *If $\|h - \widehat{h}_d\|_\infty \leq \epsilon_d$ and $\|b_k\|_\infty \leq B_k$, then*

$$|c_k(h) - c_k(\widehat{h}_d)| \leq \epsilon_d B_k.$$

If h is L_h -Lipschitz and $\widehat{h}_d$ is the regular-midpoint piecewise-constant reconstruction, then $\epsilon_d \leq L_h/(2d)$. Consequently the error of the frequency-penalized coefficient is at most

$$\frac{L_h B_k}{2d(1 + \kappa k)}.$$

Linear interpolation from a reference grid of size d_{ref} has the analogous bound $L_h/(2d_{\text{ref}})$ before deterministic rounding.

The bound says that nominal dimension disappears once the grid is fine enough to reconstruct the same profile. It justifies comparisons across discretizations of one ordered rule; it does not turn an unordered vector into a function. The proof is the integral triangle inequality followed by the nearest-node or linear-interpolation error.

5.2. Why ten diverse profiles cover the library

For any coordinate u on the finite library, define the covering radius of $A \subset \mathcal{L}$ by

$$r(A; u) = \max_{h \in \mathcal{L}} \min_{a \in A} \|u(h) - u(a)\|_2.$$

THEOREM 1 (Farthest-first finite k -center bound). *Let A_k be the k centers produced by Gonzalez farthest-first from any declared first center, and let $r_k^\star$ be the smallest covering radius attainable by any k library members. Then*

$$r(A_k; u) \leq 2r_k^\star.$$

The standard witness proof uses the k centers and one farthest unselected point. Two of these $k + 1$ pairwise-separated witnesses must share an optimal cluster. The implementation emits this witness certificate for every design. For the implemented $\eta = (z, r^g, r^f)$, Euclidean projection gives

$$r(A_k; z) \leq r(A_k; \eta) \leq 2r_k^*(\eta). \quad (14)$$

Thus source ranks can steer selection without erasing structural diversity. Equation (14) gives a checkable cover in profile space; it does not claim that source ranks have the same meaning on the target.

5.3. What finite source samples can rank

Let s_{qj} denote either a true source objective statistic or a true source chance-margin statistic and $\widehat{s}_{qj}$ its replicated estimate.

PROPOSITION 2 (Separated-rank recovery). *Suppose $|\widehat{s}_{qj} - s_{qj}| \leq r_q$ simultaneously over a finite source library. Every pair satisfying $|s_{qj} - s_{q\ell}| > 2r_q$ has the correct estimated order.*

For the Gaussian simulator used in the registered synthetic study, let $R \geq 2$, $\nu = R - 1$, and let $q_{\nu, p}$ be the p quantile of χ_ν^2 . For one profile with true mean μ and scale σ , define

$$r_\mu(\delta_\mu) = \sigma \sqrt{\frac{2 \log(2/\delta_\mu)}{R}}, \quad (15)$$

$$a_\nu(\delta) = 1 - \sqrt{q_{\nu, \delta/2}/\nu}, \quad (16)$$

$$b_\nu(\delta) = \sqrt{q_{\nu, 1-\delta/2}/\nu - 1}, \quad (17)$$

$$r_\sigma(\delta_\sigma) = \sigma \max\{a_\nu(\delta_\sigma), b_\nu(\delta_\sigma)\}. \quad (18)$$

The normal mean tail and Cochran identity $(R - 1)S^2/\sigma^2 \sim \chi_{R-1}^2$ imply, with probability at least $1 - \delta_\mu - \delta_\sigma$,

$$|\widehat{m} - m| \leq r_\mu(\delta_\mu) + z_{1-\alpha} r_\sigma(\delta_\sigma), \quad \sigma \geq 10^{-3}. \quad (19)$$

The variance floor is harmless here because $a \mapsto \max\{a, 10^{-3}\}$ is one-Lipschitz. Allocating the two tail probabilities across the finite set of tasks, profiles, and score heads gives the simultaneous radius required by Proposition 2. This result deliberately leaves nearly tied profiles unresolved: three replications cannot reliably order them. The sensitivity experiment therefore varies replications and library size instead of treating the source order as known. The normal and chi-square sampling laws supply the probability bounds (Cochran 1934); Lean checks the finite variance identity, the effect of the variance floor, the union of bad events, and the final rank implication.

5.4. From uncertain ranks to the selected design

The source information has two distinct roles: it selects the first profile and perturbs the covering metric. The following finite statement separates them. For a score head on a J -profile source library, $J \geq 2$, let $U_{qj}(r_q)$ count other profiles whose true scores differ from s_{qj} by at most $2r_q$.

PROPOSITION 3 (Rank and design stability). *On the simultaneous score-error event of Proposition 2, the normalized average-tie rank satisfies*

$$|\widehat{r}_{qj} - r_{qj}| \leq U_{qj}(r_q)/(J-1).$$

Let ϵ_g, ϵ_f bound the resulting aggregate rank errors under nonnegative source weights summing to one. For $\omega \in [0, 1]$ and $b_j = \omega r_j^g + (1 - \omega)r_j^f$, put $\epsilon_b = \omega\epsilon_g + (1 - \omega)\epsilon_f$ and $\epsilon_{\text{aug}} = (\epsilon_g^2 + \epsilon_f^2)^{1/2}$. The estimated first center $\widehat{j}$ satisfies

$$b_{\widehat{j}} - \min_j b_j \leq 2\epsilon_b.$$

It equals the unique population-score first center if the score gap exceeds $2\epsilon_b$. Conditional on this equality, the complete farthest-first sequence is unchanged if, at every remaining selection step, the population distance-to-selected-set gap between its winner and runner-up exceeds $4\epsilon_{\text{aug}}$.

The proof in the Online Supplement counts unresolved pairwise comparisons and propagates coordinate perturbations through Euclidean distances. The first-center guarantee concerns source score quality; the sequence guarantee concerns sensitivity to source noise. Both permit nearly tied rankings and cover choices to remain uncertain. When the sequence is unchanged, its mapped target design and paired evaluation law are unchanged as well. Target benefit still requires the alignment condition below. These analytic implications supplement the existing Lean-checked components.

The Online Supplement examines a further condition linking first-center score uncertainty to target safety. Its hidden-truth diagnostic satisfies the sufficient condition on 0/240 confirmation tasks. It therefore remains a boundary statement, rather than an explanation of the observed gain.

5.5. When library coverage reaches a target safe region

Let $m_q(h)$ be the target chance margin, with $m_q(h) \leq 0$ implying feasibility. Let $z_q^\star$ be an ideal target structural coordinate and $\widehat{z}$ the implemented structural projection of η . The following statement exposes the assumption that source and target share useful profile geometry; it imposes no target interpretation on (r^g, r^f) . Write $d_q(h, h') = \|z_q^\star(h) - z_q^\star(h')\|_2$.

ASSUMPTION 1 (Aligned profile geometry). *There exist a target safe-center profile $h_q^\star$, a library member h_s , and constants $\gamma_q, \Delta_q, \epsilon_\eta, L_q \geq 0$ such that*

$$\begin{aligned} m_q(h_q^\star) &\leq -\gamma_q, \\ d_q(h_s, h_q^\star) &\leq \Delta_q, \\ \|\widehat{z}(h) - z_q^\star(h)\|_2 &\leq \epsilon_\eta, \\ |m_q(h) - m_q(h')| &\leq L_q d_q(h, h'). \end{aligned}$$

The third inequality holds for every profile used below.

THEOREM 2 (Aligned atlas coverage). *Let $\mathcal{A}_{n_0}$ cover $\mathcal{L}$ within radius $r_{n_0}^\eta$ in augmented η . By (14) it covers the same support within that radius in $\widehat{z}$. Under Assumption 1, if*

$$L_q(r_{n_0}^\eta + \Delta_q + 2\epsilon_\eta) \leq \gamma_q, \quad (20)$$

then $\mathcal{A}_{n_0}$ contains a target-feasible policy.

The condition has a direct reading. The ten profiles must cover the library closely; the source-supported shape must not move too far on the target; the implemented coordinate must approximate the target-relevant geometry; and the safe region must be deep enough to absorb all three errors. Frequency shifts, irregular grids, and misspecification increase Δ_q or ϵ_η , while a shallow feasible region decreases γ_q . In those cases the theorem stops, as it should. The complete triangle-inequality proof appears in the Online Supplement.

Figure 2 gives the geometric interpretation. The symbol ϵ_η denotes error between the implemented projected structural coordinate $\widehat{z}$ and the ideal target coordinate $z_q^\star$; it does not transfer the source-rank components (r^g, r^f) to the target. The circle labeled r_{n_0} depicts the structural-space upper bound obtained by projecting the augmented-coordinate cover $r_{n_0}^\eta$.

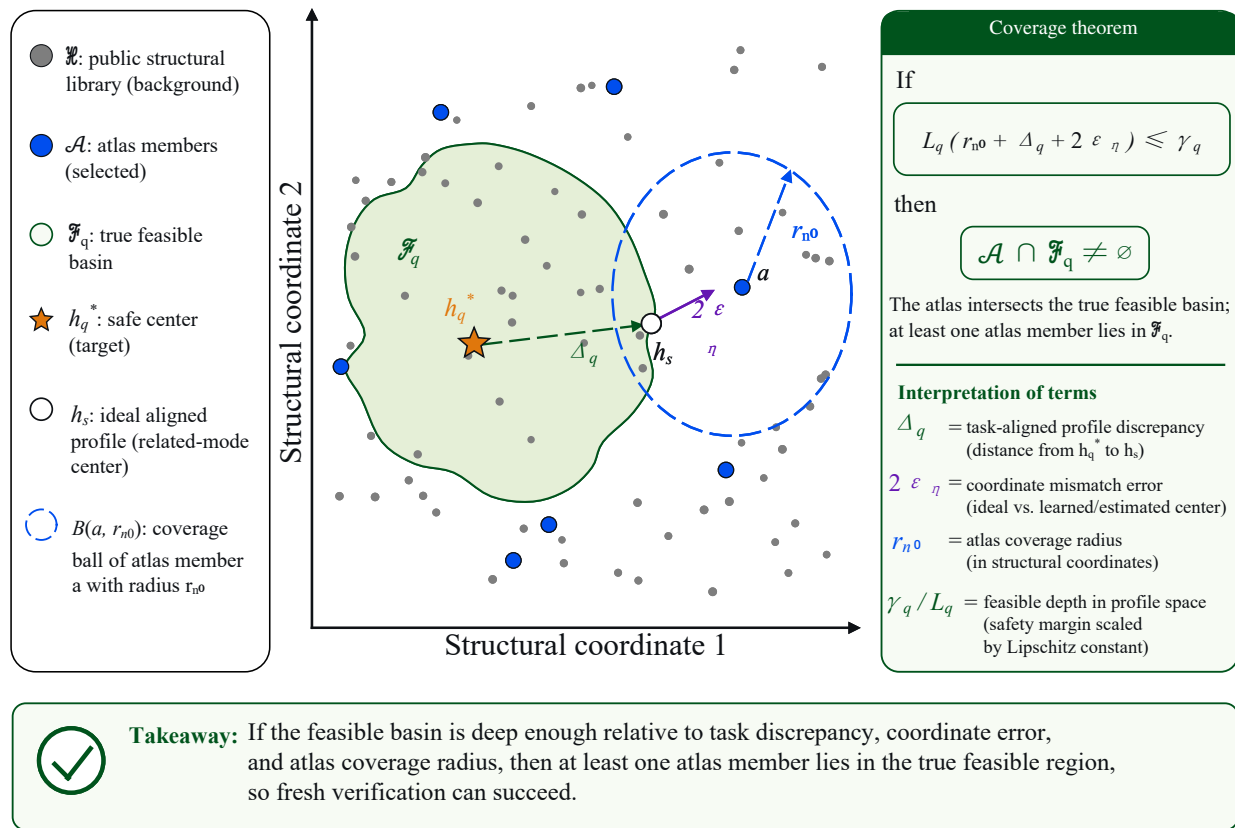


Figure 2 Schematic geometry of atlas coverage in structural profile space. Gray points represent the public profile library and blue points the selected atlas. The target has an unknown feasible basin $\mathcal{F}_q$ containing a safe center h_q^* . A source-supported library profile h_s lies within task discrepancy Δ_q of the safe center; projected coordinate mismatch contributes at most $2\epsilon_\eta$; and an atlas member a covers h_s within the projected upper bound r_{n0} . Hence $L_q(r_{n0} + \Delta_q + 2\epsilon_\eta) \leq \gamma_q$ implies $\mathcal{A} \cap \mathcal{F}_q \neq \emptyset$. The diagram illustrates the coverage theorem and is not an empirical two-dimensional projection of a target instance.

The preceding theorem concerns one target. To estimate how often the selected design succeeds over the declared task population, we next treat a design hit as a bounded random outcome. The confirmatory generator uses eight fixed, equally weighted regimes, not one pooled IID law. At a fixed resolution d , let regime s have n_s independent held-out task seeds, weight w_s with $\sum_s w_s = 1$, and hit indicators $H_{si} \in [0, 1]$. Define the task-law hit rate $p = \sum_s w_s \mathbb{E}[H_s]$, its stratified estimate $\hat{p} = \sum_s (w_s / n_s) \sum_{i=1}^{n_s} H_{si}$, and $v_w = \sum_s w_s^2 / n_s$. Weighted Hoeffding concentration, which does not require identical distributions across strata, gives with probability at least $1 - \delta$,

$$p \geq \hat{p} - \sqrt{\frac{v_w \log(2/\delta)}{2}}. \quad (21)$$

Here $w_s = 1/8$ and $n_s = 20$. The deterministic component `task_seed mod 5` is an induced latent category whose realized counts need not be balanced; independence and boundedness, rather than identical distributions, are sufficient. The same 160 latent task seeds are crossed with all three resolutions, so (21) is applied

separately at each d ; the pooled 480-cell rate is descriptive. The probability refers only to a prospective repetition of the registered stratified generator and requires the atlas to be frozen before held-out tasks. It is not a guarantee for arbitrary future domains.

5.6. Independent terminal safety

Fix a shortlist before verification. Candidate j receives v_j fresh iid Bernoulli indicators $Z_{jr} = 1\{Y_{jr}^g \leq \tau\}$ and is certified only when $\sum_r Z_{jr} = v_j$ and

$$p_0^{v_j} \leq \delta_j, \quad p_0 = 1 - \alpha, \quad (22)$$

where $\delta_j = \delta/K_s$ in the primary protocol.

THEOREM 3 (Candidate and familywise validity). *For any fixed unsafe candidate with true feasibility probability $p_j < p_0$,*

$$\mathbb{P}\{\text{candidate } j \text{ is certified}\} = p_j^{v_j} \leq p_0^{v_j} \leq \delta_j.$$

For a frozen shortlist, regardless of dependence among the candidate tests,

$$\mathbb{P}\{\text{an unsafe policy is deployed}\} \leq \sum_{j=1}^{K_s} \delta_j \leq \delta.$$

Sequentially stopping at the first certificate does not alter this union-bound argument because the shortlist and all budgets are fixed before testing.

For $p_0 = .95$, $K_s = 3$, and $\delta = .05$, the minimum all-success budget is 80 per candidate. Its power at true feasibility probability p is p^{80} ; this is only 0.017 at $p = .95$. Validity therefore does not imply that the test will often certify, and Section 7 reports both validity and power. The exact Clopper–Pearson lower bound is used only for the preregistered sensitivity calculation (Clopper and Pearson 1934). A complete proof of the familywise statement appears in the Online Supplement.

5.7. Human-readable and machine-checked proofs

The Online Supplement gives ordinary mathematical proofs of every named result in this section, the task-law bound, the finite-atlas limitation, and the call accounting identities. A bidirectional closure table maps each manuscript original statement to exact Lean declarations in the frozen paper-facing interface. These declarations build in Lean 4 without `sorry`, `admit`, or a project-defined `axiom`. Proposition 3 and the supplementary first-center margin-transfer proposition have analytic proofs and are outside this original interface. Machine checking confirms that the formalized conclusions follow from the stated premises. It cannot establish that Assumption 1 holds in Energy or in an unspecified future domain, nor can it replace the declared Gaussian, bounded-task, and binomial sampling laws; those remain scientific premises.

6. Experimental Design

The original confirmatory protocol was fixed before observing its outcomes. A compact registry records its method version, cells, failures, and analysis endpoints. Supplementary comparisons added after manuscript review reuse those target seeds and are identified separately below. Every implementable method sees target truth only after its deployment decision, when truth is used to compute evaluation metrics.

6.1. Randomized ordered-profile stress test

The primary study uses eight fixed, equally weighted regimes. Within each regime, independent deterministic seed streams generate held-out problem instances. Table 2 separates grid resolution from the complexity of the underlying policy shape.

For each regime, we draw 20 independently seeded latent target tasks and cross the same seeds with $d \in \{200, 1000, 10000\}$. This gives 160 independent latent task seeds and 480 correlated task–resolution cells per design. Each cell in the original frozen matrix is paired across six designs: source-scored, unstandardized DCT maximin, random low frequency, natural three-block, raw Sobol, and an explicitly labeled finite-library upper bound. The primary implementable matrix has $8 \times 3 \times 20 \times 5 = 2400$ cells; including the oracle gives 2880. Every design uses $n_0 = N = 10$, so the primary comparison is a pure initial-design intervention. The source-scored arm pays 384 source calls; source-free arms pay none. All use the same independent shortlist verifier.

Supplementary matched comparisons. We added standardized DCT maximin and the source-greedy portfolio on the same 160 latent tasks at all three resolutions. These controls use $n_0 = N = 10$, the original target noise streams, and the primary (80, 80, 80) verifier. Standardized DCT matches the source-scored atlas’s structural scaling; the portfolio also matches its 384-call source archive. The standardized control at $d = 1000$ was first evaluated during manuscript review and is reused unchanged; the remaining runs followed the recorded supplementary rules. All rules were chosen after the original target results had been examined. They isolate the comparison confound and selection-rule choice on reused tasks; they are not a new independent confirmation. The original matrix and its results remain frozen.

For each synthetic target regime, the two supplied source tasks are generated from that same regime with the role set to source. This is a regime-matched transfer experiment, not a source-retrieval experiment. Algorithms receive neither target outcomes nor hidden target geometry, but the experimenter has already supplied a related archive. The Energy region holdout is the explicit archive-mismatch control.

The primary chance level is $\alpha = .05$, and task safe mass is calibrated to .08 on a separate fixed library that is not used for design. Regret is measured against the best feasible policy in that evaluation library. The $d = 10000$ cells refine the grid while holding the underlying problem fixed unless the regime explicitly changes rank.

First-center attribution and new-family confirmation. We subsequently fixed a factorial comparison of the structural medoid and source-best first center, crossed with structural-only and augmented metrics. The medoid was held fixed across the two metrics. A fifth design used the source-best center with Voronoi-weighted raw-profile L^2 distance on the same 128 reference nodes. These diagnostic comparisons reused the original tasks; identical ordered designs reused their original outcomes with the proper source cost. A separate local prospective protocol then fixed three new family seeds, three corresponding public libraries and source archives, and ten new tasks per mechanism and family at $d = 1000$. Its 240 latent tasks were compared across six designs, including the same-source portfolio. The primary contrast was source scoring minus standardized generic maximin in certified true-feasible deployment. Intervals resampled tasks within the 24 fixed family–mechanism strata, and each family was reported separately. Library sizes 32 and 128 were additional fixed sensitivities on the first new family; they change support composition as well as cardinality. The archive, selection rule and verifier were not tuned after these new outcomes.

6.2. Sensitivity and information controls

An 8640-cell one-factor-at-a-time matrix varies active rank, α , safe mass, n_0 , number of source tasks, profiles per task, source replications, retained frequency, frequency penalty, source-rank metric weights, and the first-center safety weight. These experiments describe sensitivity; they are not a hyperparameter search performed after seeing the primary results. A separate 1280-cell matrix crosses declared versus schema-blind profile mapping with equal versus descriptor-conditioned source weights. The latter weighting is excluded from the primary method regardless of its result.

6.3. Cost and optimizer controls

The equal-preverification-cost matrix compares 384 source plus 10 target calls with 394 target-only calls for each source-free design. The same independent verifier is applied afterward and all-in realized calls include its cost.

Target-only functional SCBO searches eight nonconstant cosine coefficients and one level parameter (nine optimization variables) with the published constrained trust-region posterior-sampling backend (Balandat et al. 2020, Eriksson and Poloczek 2021). We test coefficient counts 4, 8, and 16 at $N = 13$, and the primary 8-coefficient version at $N = 394$. One $N = 394$ task in the irregular-grid regime returned duplicate canonical posterior samples; it remains a failed, unsuccessful cell.

Eight native transfer pipelines (Safe F-PACOH, RGPE, stacked transfer GP, multi-task GP, FSBO, HyperBO, MetaBO, and MALIBO) receive the same 384-call source archive, $n_0 = 10$, and $N = 13$, but retain their own prior, representation, initialization, and acquisition logic. This is an end-to-end transfer comparison on three fixed families with 20 algorithmic seeds per family. All eight pipelines share the historical v69 terminal protocol: candidate budgets (80, 128, 128) and an 8-call incumbent objective comparison. This protocol differs from the primary (80, 80, 80) verifier; native results are interpreted within their own matrix.

6.4. Electricity-reserve external comparison

The external study uses the Open Power System Data time-series release (Open Power System Data 2020). A 1000-point policy maps observable forecast stress to reserve responses over a 168-hour service window. The corrected Energy V5 model gives every policy the same initial SOC, 50% of capacity, charges all subsequent grid purchases, and records terminal energy. Reserve adjustment and forecast-error balancing share one hourly converter budget: the sum of external charging and discharging energy cannot exceed the power capacity times one hour. Reserve adjustment is executed first and balancing uses the remaining budget. Its objective is finite-period service cost with zero terminal salvage value, rather than a steady-state economic criterion.

All eight designs were rerun on 18 markets in five geographic regions with five algorithmic seeds per market, giving 720 cells. Each uses $n_0 = 10$, $N = 13$, and the same (80, 80, 80) verifier. One source archive is frozen for each excluded geographic region before target evaluation, with four external source markets, 64 profiles and three replications (768 calls). Its source outcomes and selected designs are shared across the region's target markets and source-based comparisons. The data splits remain search in 2017, audit in 2018 and verification in 2019. V2–V4 results are preserved as historical evidence; V5 supplies the current external comparison.

Fresh verification windows are sampled from the held-out empirical calendar. After each decision, four chronological blocks and physically nonoverlapping windows provide descriptive stability checks. Actual reuse accounting charges one archive per excluded region and sums target costs across its observed markets; five seeds estimate repeatability within each market and do not count as five deployments. This corrected study reuses previously examined data, so it is distinct from the prospective synthetic confirmation.

6.5. Inference and failure accounting

For randomized profiles, a latent task seed within a fixed regime and resolution is the generalization unit. Supplementary paired bootstrap intervals resample task seeds within each fixed regime and retain the equal regime weights. Task-law concentration is applied separately at each resolution with regimes as fixed strata. Rates pooled over all three resolutions describe 480 task–resolution cells but are not treated as 480 independent tasks. Simulation seeds within a fixed external market measure algorithmic repeatability, while geographic regions are the conservative Energy comparison units. We do not pool seeds as if they were independent domains. Exact sign tests are Holm-adjusted within their registered families, but effect sizes and intervals receive priority over very small p -values. Every timeout, duplicate-sampling failure, and false certificate remains in its denominator.

Primary endpoints are feasible coverage, certified true-feasible deployment, false certification, and penalized loss. We additionally report source, search, verification, unamortized all-in, and $M = 20$ amortized calls; conditional regret is secondary.

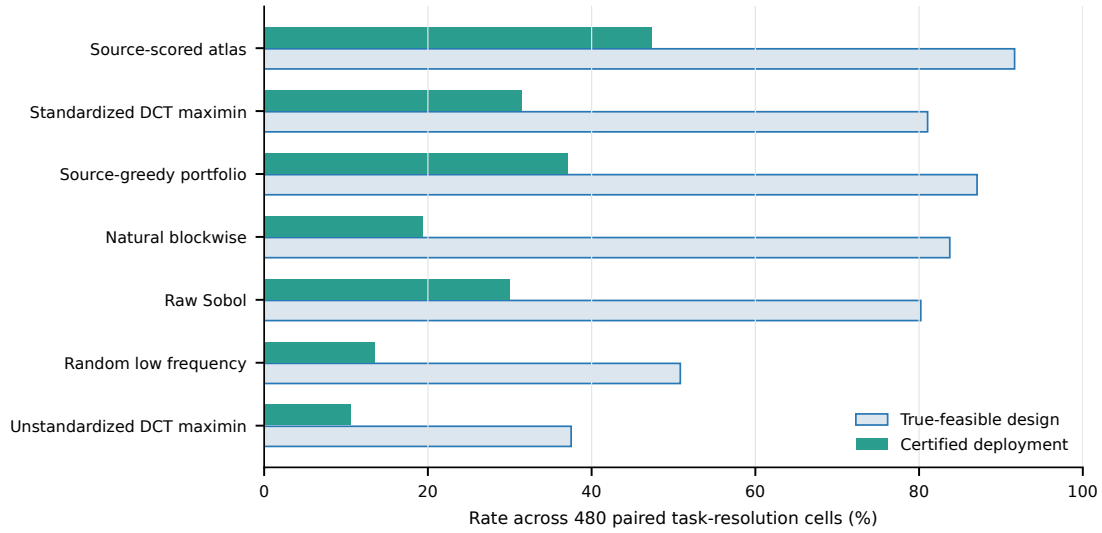


Figure 3 Aggregate feasible coverage and independently certified deployment. Bars aggregate task-resolution cells; inferential comparisons are paired within resolution and do not treat the three grids as independent tasks.

7. Results

7.1. Aggregate randomized-task performance

Table 3 and Figure 3 report the primary randomized task law. The source-scored design contains a truly feasible policy in 91.7% of task-resolution cells and yields a certified true-feasible deployment in 47.3%. In the supplementary comparisons, standardized DCT maximin reaches 81.0% and 31.5%, and the source-greedy portfolio reaches 87.1% and 37.1%. Penalized losses are 0.163, 0.274, and 0.178, respectively. Source scoring retains an aggregate advantage over matched structural scaling and a simple alternative source-selection rule.

The original unstandardized DCT control reaches only 37.5% coverage and 10.6% certification. Its difference from the standardized control shows that the original comparison combined source scoring with a change of metric; it cannot isolate the value of source outcomes. Raw Sobol reaches 80.2% coverage and 30.0% certification with penalized loss 0.337.

Paired source-minus-standardized-DCT certification differences are 0.163 (stratified bootstrap 95% CI [.119, .206]) at $d = 200$ and 0.156 ([.113, .200]) at both $d = 1000$ and $d = 10000$. Against the source-greedy portfolio they are 0.113 ([.044, .181]) at $d = 200$ and $d = 1000$, and 0.081 ([.012, .150]) at $d = 10000$. Each interval resamples the 20 paired tasks within each of the eight fixed regimes; the resolutions are analyzed separately. The Online Supplement reports paired wins, losses, and ties. These supplementary estimates describe corrected comparisons on reused tasks. Grid refinement preserves the aggregate advantage under this task law.

7.2. First-center attribution and prospective confirmation

The diagnostic factorial comparison isolates the source-selected first center. At $d = 1000$, source-best with structural distance alone certifies 75/160 true-feasible deployments, compared with 74/160 for the full atlas and 49/160 for either medoid design. Appending ranks to a fixed medoid does not recover the gain. Structural-only and augmented source-best designs have the same feasible coverage at every resolution; the certification difference is one task per resolution. The Online Supplement gives the complete factorial and raw-profile-distance controls.

Table 4 reports the separate prospective experiment on 240 new latent tasks, three new source families and three new public libraries. The atlas certifies 56/240 true-feasible deployments, compared with 24/240 for standardized generic maximin: a paired difference of 13.3 percentage points (95% stratified bootstrap interval [9.6, 17.1]). The three family-specific differences are 26.3, 10.0 and 3.8 points. Source-best with structural distance alone certifies 54/240, so the rank-distance increment is only 0.8 points ([0, 2.1]). The atlas exceeds the matched raw- L^2 design by 4.2 points ([1.7, 6.7]), supporting a limited role for the structural metric.

The same-source portfolio certifies 49/240; the atlas-minus-portfolio difference is 2.9 points ([−2.1, 7.9]). Its feasible coverage is 212/240, compared with 155/240 for the atlas, and its mean penalized loss is 0.156 versus 0.414. Each of the six designs records one false certificate. Thus the new experiment confirms a source-initialization advantage over generic maximin, but does not confirm atlas superiority over a portfolio. The postdecision first-center audit finds 106/240 truly feasible source-best centers and 83/240 medoids; it is a diagnostic, not an input to selection. The predefined 32- and 128-profile sensitivities preserve positive paired certification differences on the first new family (15.0 and 13.8 points), without establishing a monotone library-size effect.

7.3. Does structural coverage improve the remaining nine policies?

A postdecision diagnostic fixes the same source-selected first policy and replaces the other nine by a uniform sample without replacement from the remaining 63 library members. It retains all 240 tasks, original target observation streams, ten-point budget, shortlist and verifier. The random remainder certifies 53/240 true-feasible deployments, versus 54/240 for source-first structural coverage and 56/240 for the full atlas. The paired certification differences are 0.4 percentage points (95% stratified bootstrap interval [−3.3, 4.2]) and 1.3 points ([−2.5, 5.0]), respectively. This diagnostic does not establish a certification advantage of structural coverage over random remainder selection. Initial feasible coverage is 167/240 for the random remainder and 155/240 for either geometric design.

A fixed-first-center portfolio control changes ten ordered designs and reuses 230 identical ordered designs. It retains 212/240 feasible coverage and 49/240 true certificates. A decision-chain reconstruction explains why these endpoints disagree: the portfolio's three-member shortlist contains a truly feasible policy in 177

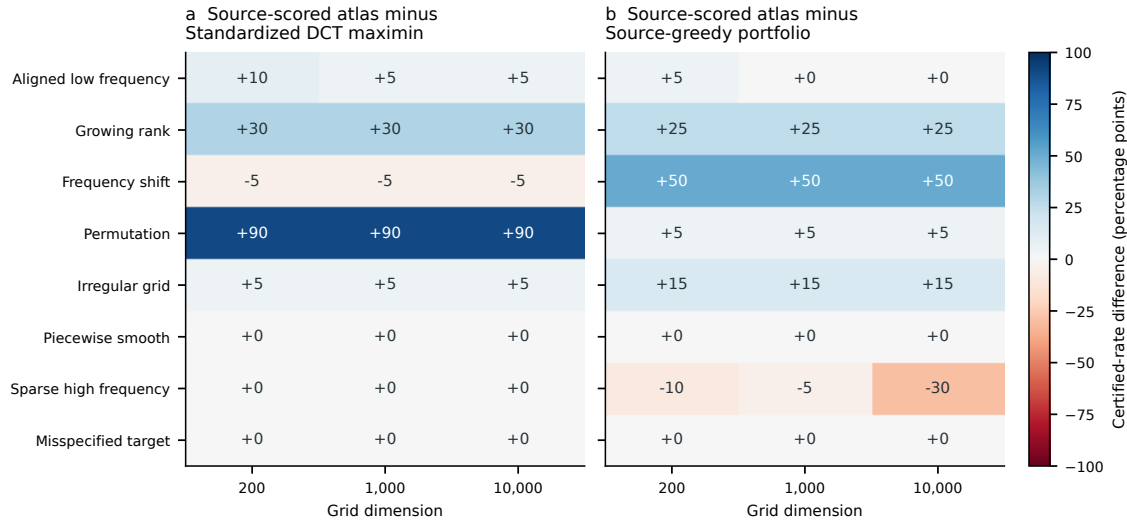


Figure 4 Source-scored atlas minus each supplementary control in certified success, by regime and grid resolution. Entries are percentage-point differences over 20 paired tasks.

tasks, losing 35 design hits, whereas the atlas retains all 155 hits. Conditional on the original noisy shortlists, analytic expectations over fresh verification streams are 54.07 true certificates for the fixed-first portfolio and 49.72 for the atlas. These expectations explain verification variability; they do not replace observed outcomes. The evidence supports source initialization over generic selection, while leaving the incremental value of the covering rule unresolved. The Online Supplement reports this later diagnostic separately from the prospective confirmation.

7.4. Where source scoring fails

The aggregate advantage is not uniform. Table 5 and Figure 4 preserve every adverse regime. Raw Sobol outperforms source scoring in aligned low frequency and irregular grids; natural structure is slightly better under sparse high-frequency activity; and neither method certifies useful policies under full misspecification. The matched comparisons localize that advantage. At $d = 1000$, 24 of the 25 net certificates gained over standardized DCT come from coordinate permutation (18) and growing rank (6). Piecewise smoothness ties at 19/20, and frequency shift slightly favors standardized DCT, 13/20 versus 12/20. Against the portfolio, the largest gains occur in frequency shift and growing rank. In the sparse-high-frequency regime its feasible coverage falls from 20/20 tasks at $d = 200$ to 0/20 at $d = 10000$, despite unchanged target budget. The portfolio covers 20/20 and certifies 6/20 of these $d = 10000$ tasks, whereas both source-scored and standardized maximin cover and certify none. Geometric diversity is useful on the mixture without being uniformly preferable to source-based greedy selection.

The source-scored and standardized DCT arms record no false certificates in 480 task-resolution cells each. Unstandardized DCT, the source-greedy portfolio, and raw Sobol record three, three, and two, respectively.

Target-only functional SCBO at $N = 13$ certifies 15 of 480 primary cells with three false certificates, all in the frequency-support-shift regime. Increasing its cosine rank does not remove the general coverage deficit.

The aligned-low-frequency reversal has a concrete geometric explanation. The supplementary geometry diagnostic reconstructs each fixed design and accesses the hidden target safe center only afterward. Across the 600 design points per arm, raw Sobol has median latent-center distance 0.0343 and median non-DC low-frequency energy 0.0258, compared with 0.0813 and 0.2292 for source scoring. Its pointwise true-feasible rate is 94.7%, versus 10.0% for both source and the original unstandardized DCT. High-dimensional independent coordinates average toward a near-constant profile with DC coefficient .5; the aligned generator's safe center happens to occupy that region. This after-the-fact diagnostic explains the reversal, but it uses hidden geometry and cannot be used to choose a method or change an outcome.

7.5. Schema and sensitivity

Declared profile ordering is valuable information. With domain-blind source weights, removing the declared schema reduces certified success from 46.2% to 35.0% (Table 6). Descriptor-conditioned weighting reduces both declared- and blind-schema performance and is not used in the final method. Source scoring therefore does not need the target family label, but it does rely on an interpretable ordered profile map.

Sensitivity is nonmonotone, which argues against treating source volume as a free advantage. Raising profiles per source task from 64 to 128 reduces certified success from 46.2% to 38.8%; using one, three, or five replications gives 45.0%, 46.2%, and 45.6%. Four source tasks give 45.0%, not more than two. Retaining 16 frequencies reaches 48.1%, while four reaches 43.1%. Increasing n_0 from 10 to 20 yields only 46.9% certification, while $n_0 = 5$ yields 42.5%. The first-center safety weight is more consequential: .25 reduces certification to 32.5%, whereas .5 and .75 give 46.2% and 45.0%. The Online Supplement displays the source-budget axes. The frequency-penalty sweep is algebraically redundant because column standardization cancels positive frequency scaling; it does not demonstrate robustness to an effective tuning parameter.

7.6. Cost changes the ranking

At the source method's primary budget, mean unamortized all-in calls are about 559 across the randomized law, while amortization over 20 targets reduces the source contribution from 384 to 19.2 calls per target. Standardized DCT uses 201.2 all-in calls and the portfolio uses 592.0 before amortization or 227.2 at 20 targets. Verification dominates their ten target-search calls. Figure 5 makes both accounting views explicit.

When the 384 source calls are instead given to target-only search, the source method is not best (Table 7). It certifies 46.2% of 160 tasks. Unstandardized DCT and raw Sobol each certify 53.8%, natural structure 53.1%, and target-only functional SCBO 58.1%. The latter includes one failed cell as an unsuccessful outcome. Mean realized all-in costs are already close because methods consume different verification effort. The source archive is thus an amortized multi-target investment, not a one-target total-cost win.

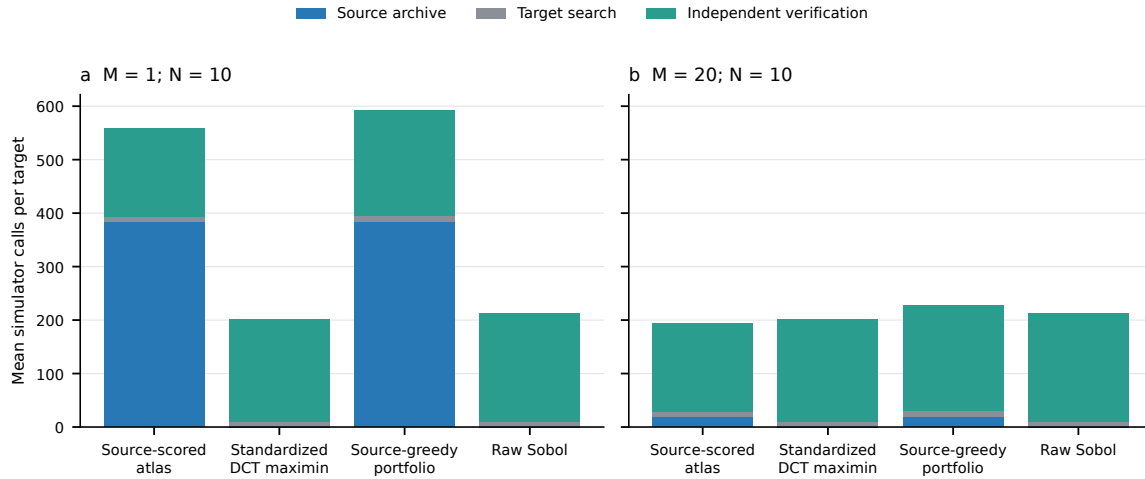


Figure 5 Mean source, target-search, and verification calls under unamortized and 20-target accounting.

Using primary mean call counts, the archive breaks even against standardized DCT at 15 targets when aggregating resolutions, or 15, 16, and 16 targets at the individual resolutions. The earlier thresholds of 7 against unstandardized DCT, 9 against natural structure, and 11 against raw Sobol depend on those comparators. Both source-based rules pay 384 archive calls; the atlas consumes less verification effort than the portfolio on average. These are descriptive cost thresholds.

Table 8 converts amortized calls into calls per true-feasible certificate. With the archive shared by 20 targets, source scoring uses 411.3 calls per certificate, versus 639.5 for standardized DCT, 612.7 for the portfolio, 1143.2 for natural structure, and 709.4 for raw Sobol. This descriptive ratio does not price unsafe deployment, abstention, or objective quality.

7.7. External Energy result

Energy does not establish a robust certification or archive cost advantage under the corrected physical model. Table 9 reports 57/90 certified source runs and 54/90 for unstandardized DCT maximin. Standardized DCT, the same-source portfolio, random low frequency, natural constants, raw Sobol and functional SCBO certify 49, 59, 56, 52, 43 and 51 runs. All eight designs record zero false certificates. The source-minus-unstandardized-DCT region-balanced difference is 0.027 (descriptive bootstrap 95% interval $[-.040, .120]$). Against standardized DCT it is 0.060 $[-.000, .140]$, and against the portfolio -0.013 $[-.040, .000]$. The small aggregate certification gain is unresolved across geographic regions and does not establish superiority over the portfolio.

Of 421 certificates in the corrected eight-arm matrix, 374 satisfy the chronological-block criterion, all 421 satisfy the nonoverlapping-window criterion, and 374 satisfy both. These are descriptive calendar-stability results. The Online Supplement reports each design separately.

Actual archive reuse across the 18 markets does not rescue the external cost comparison. Averaging the five repeats within each market, the source arm consumes 6842 cumulative calls, including 3840 source calls charged once across five regional archives, and yields 11.4 expected certified deployments. Unstandardized DCT consumes 3258 calls and yields 10.8. Calls per certified success are therefore 600.2 and 301.7. These observed reuse counts replace a hypothetical twenty-deployment allocation for this comparison.

7.8. Public-battery development diagnostic

A separate retrospective replay on three 2024 seasonal tasks tests whether source initialization survives simple policy controls. Source-first structural search and source-free constant-only structural search both certify 15/15 runs across five algorithm repeats per season, with zero finite-population false certificates and the same target query count. Averaging repeats within each season gives 279 target queries over the three tasks. The source design additionally pays 44,968 source-window queries; the constant-only design pays none. Testing only the source-selected first policy also certifies 15/15, using 243 target queries but retaining the same source cost.

Paid enumeration of all 154 library policies over all 73 windows per task requires 33,726 queries, below the source design's 45,247 source-inclusive queries. Restricting enumeration to the 100 constant pairs requires 21,900 queries and selects the same complete-cost optimum in each season. This diagnostic closes the development query-efficiency claim for source initialization in this case. It reports a finite-library adaptation rather than a run of the full rank-augmented atlas. The Online Supplement gives the sampling law, accounting and policy selections.

7.9. Verifier power and native transfer

The verifier is valid but intentionally conservative. At 80 all-success replications, certification power is only 1.7% at the threshold $p = .95$, 13.2% at .975, and 44.8% at .99 (Figure 6). Therefore the gap between 91.7% feasible coverage and 47.3% certified deployment is expected; an empty certificate set is not evidence that search found no feasible point. A Clopper–Pearson threshold rule is reported as sensitivity, not substituted after observing outcomes.

In the native transfer matrix, RGPE certifies 30/60 tasks and MetaBO 28/60; the other six methods range from 0/60 to 14/60 (Online Supplement, subsection “Native end-to-end transfer”). All eight certify 0/20 Queue tasks. These results show that ordinary end-to-end transfer remains difficult, but they do not establish a broad SOTA ranking from three historical synthetic families. Their v69 verification protocol differs from the primary exact-binomial protocol, so certification totals should not be compared across these matrices.

Finally, cumulative factor-HVD substantially improves held-out log-variance RMSE and variance-shape correlation relative to pooled variance, yet does not improve feasible recovery, regret, false certification, or verification cost under matched designs. It is therefore retained only as a calibration diagnostic in the Online Supplement, Section EC.3, not as a main optimization contribution.

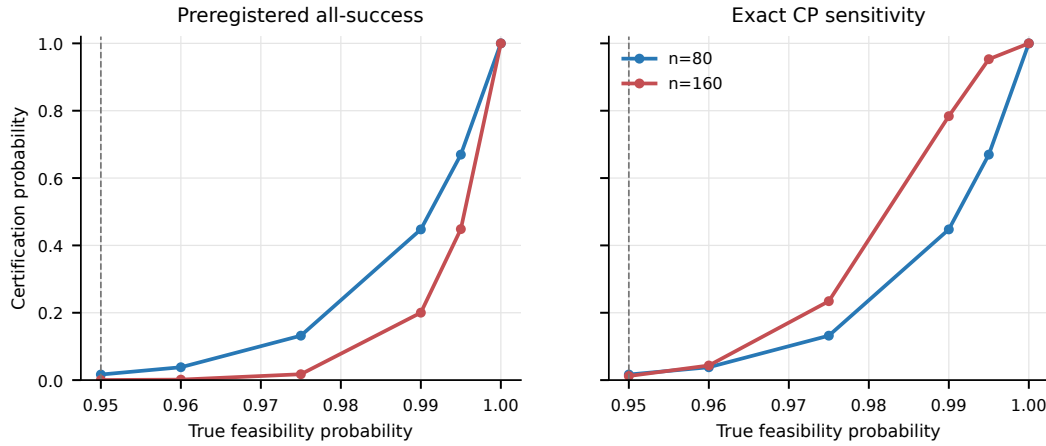


Figure 6 Exact fixed-budget certification power. The preregistered all-success rule is familywise valid but weak near the chance threshold.

8. Discussion

8.1. What the evidence supports

Source outcomes improve certification relative to standardized generic maximin on both the original mixture and new family instances. The factorial comparison identifies the source-best first center as the main effective component. Rank augmentation has little additional effect at the frozen weights. Structural coordinates improve certification over matched raw profile distance, but geometric coverage is not uniformly preferable to source-based portfolio selection: the prospective portfolio comparison does not reproduce the original aggregate certification advantage and favors the portfolio in coverage and penalized loss. Which mechanism benefits also changes with the family and public library. In particular, coordinate permutation contributes to the original gain but reverses in the new study.

Fixing the source-selected first policy also exposes the distinction between design quality and deployment certification. A random nine-policy remainder produces similar certification to structural coverage. The portfolio has better feasible coverage but loses feasible candidates during noisy shortlist selection. Certification therefore measures the whole selection-and-testing pipeline; it does not by itself establish a superior initial point set.

This interpretation differs from claiming a generic high-dimensional optimizer. The raw vector may contain 10^4 coordinates, but the decision is an ordered profile and the library has finite structural complexity. A controller may still require a fine lookup table, so profile-grid refinement is operationally relevant. It is not equivalent to adding 10^4 unrelated decision variables. The proper benefit is invariance to representation resolution under shared profile structure.

8.2. What Energy teaches us

Energy shows that a small aggregate certification gain need not repay the source archive. The corrected V5 model gives an unresolved region-balanced gain over unstandardized maximin and about twice its calls per certificate. Its finite seven-day service objective uses a common initial inventory, shared hourly power and zero terminal salvage value. This is a practical boundary on archive use. The public-battery diagnostic strengthens this lesson: an outcome-free constant-policy search matches certification, and paid target enumeration fits within the source-inclusive query budget. For these development tasks, the archive is unnecessary to obtain the reported certification outcome.

The practical implication is to compare source-scored and outcome-free designs on held-out tasks with the same ordered semantics before committing to archive reuse. Energy alone cannot identify a universally preferable scaling or a rule for detecting transfer failure before target outcomes.

The current method begins after this assessment: a source archive has already been supplied, and the algorithm ranks profiles inside it. It does not retrieve related tasks from an unrestricted historical database. Regime-matched synthetic sources therefore identify the value of scoring a relevant archive, whereas Energy shows an archive investment that does not pay off over the observed deployments. Source retrieval and its own data-acquisition cost remain separate research and operational decisions.

8.3. When the source archive pays for itself

Ten target calls do not beat 394 target calls for a single target. The source archive becomes economically plausible only when it is reused. Let $C_0 = N_0 + V_0$ be the expected target-only calls per deployment and $C_A = N_A + V_A$ the source-design calls excluding the archive. The call-count break-even number of deployments is

$$M_{\text{break}} = \left\lceil \frac{S}{C_0 - C_A} \right\rceil, \quad C_0 > C_A. \quad (23)$$

In the randomized primary experiment, the matched standardized control gives a threshold of 15 targets using aggregate mean calls, and 15–16 targets when each resolution is analyzed separately. Both source-based designs pay the same archive cost; lower verification effort favors the atlas over the portfolio on average. This arithmetic does not include differences in simulator setup, wall-clock parallelism, or policy quality, and Energy shows that paying the archive can still be inferior. It is a planning threshold, not a dominance theorem.

Table 8 also reports calls per certified deployment, but that ratio is only a descriptive planning aid. A complete operational comparison would attach decision-maker costs to unsafe deployment, abstention, and objective quality. We report those components separately rather than choose their relative value on the reader's behalf.

8.4. Search and verification answer different questions

Independent verification frequently costs an order of magnitude more than target search because it answers a different question. Search asks whether a promising policy has been found. Verification asks whether a fixed policy can be deployed with a specified false-certification bound. A target-only optimizer making the same safety claim must also pay for fresh samples.

The all-success rule is valid but weak near the threshold, which explains why feasible coverage exceeds certified deployment. Future studies should fix a more efficient exact or sequential test in advance and use block-aware verification when temporal replications are not iid.

8.5. Limitations

The task law, library, and stress axes were created within the project. The matched standardization and greedy-portfolio controls were added after reviewing the original outcomes and reuse the same target seeds. The fixed-first random-remainder and portfolio controls were likewise proposed after observing confirmation results. Their intervals describe task variation within the existing families, not independent confirmation or a guarantee of equivalence. Positive intervals quantify sampling variation within the specified task law; they do not remove development selection. The separate prospective study uses new family instances, source archives, public libraries and target tasks; its intervals describe task variation conditional on those three fixed families, rather than a population of all source libraries. Ordered semantics are a strong prior. Energy uses counterfactual storage simulations on historical data from five regions, and the native transfer study has only three historical families. The public-battery controls were proposed after observing development outcomes and use overlapping windows from one market. Their certificates refer to iid draws of window indices from a fixed uniform finite law, not independent calendar observations or future deployment. No sealed 2025 data are used. Complete costs are ranked only for policies succeeding on every window; failed full costs remain undefined. Query accounting does not establish wall-clock or monetary efficiency. The archive is assumed to have been supplied; source retrieval is unsolved. Three replications cannot order near ties, source aggregation can hide task disagreement, and a finite library can miss the safe region entirely. Regret uses a finite evaluation library rather than the unknown continuous optimum, and one functional-SCBO failure remains in the denominator. The theory is conditional on alignment, independence within each regime, and the declared replication law. Its finite geometric and sampling guarantees do not prove superiority to either comparison design. Finally, the same target seeds appear at all three grid resolutions, so 480-cell totals are descriptive rather than 480 independent tasks.

9. Conclusion

A policy stored as thousands of ordered numbers may still be one simple rule. We use source simulations to rank transparent policy shapes, farthest-first selection to keep ten starting policies diverse, and fresh

simulations to make the final safety decision. Matched comparisons and prospective new-family confirmation support a certification advantage over standardized generic selection. Factorial controls identify the source-selected first center as the main effective component, while the structural metric adds a smaller increment over raw profile distance. A later fixed-first-center diagnostic does not establish a certification advantage over nine random remainder policies. The prospective portfolio comparison favors the portfolio in feasible coverage and penalized loss; an outcome-free design has lower calls per certificate in Energy, and additional target search wins at equal single-target preverification cost. The public-battery development diagnostic is matched by a source-free constant-policy search and does not repay source queries. The supported contribution is therefore source-selected initialization under specified synthetic task laws, with finite conditional guarantees and explicit verification. Archive reuse in an operational domain requires a separate comparison against simple policies and source-inclusive target budgets.

10. Code and Data Disclosure

Code and the declared seeds generate every synthetic instance. The Energy study uses the public data cited in Section 6. An anonymous code-and-data attachment, `or_submission_revision_controls_20261002.zip`, contains source code, original and supplementary protocols, compact per-task results, the processed Energy archive, and commands to regenerate analyses, tables, figures, and PDFs. It also contains the Lean source and its pinned dependency configuration. Original protocol metadata errors are documented in a separate erratum. Historical observations are preserved; corrected Energy V5, mechanism attribution and new-family confirmation have separate protocols and per-task artifacts. The attachment also includes the postdecision remainder and first-center diagnostics, and compact public-battery replay outcomes and census summaries. Battery window-level replay requires the retained population journals, which are outside this compact attachment.

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Table 1 Frozen primary method specification.

Component	Specification
Decision class	Ordered bounded profile $h : [0, 1] \rightarrow [0, 1]$ on a declared grid
Library	64 outcome-free profiles; 128 reference nodes; library seed 20261799
Source archive	2 tasks $\times$ 64 profiles $\times$ 3 replications = 384 calls
Profile coordinate	Frequencies 0:8, diagonal squares, 18 standardized dimensions
Source score	Equal-task average of objective and Gaussian chance-margin percentile ranks
Selection	First center $.5r^g + .5r^f$; nine Gonzalez farthest-first centers
Metric weights	Profile coordinates standardized; safety and objective rank weights both 1
Target map	Linear interpolation to declared nodes; deterministic integer rounding
Target information	Grid, bounds, dimension, and declared ordering; no outcomes or oracle quantities

Table 2 Registered stress regimes. The target generator hides its basis, center, safe radius, and task seed from every algorithm.

Regime	Default rank	Principal departure
Aligned low frequency	8	Source and target share low-frequency support
Growing effective rank	24	More active coefficients at fixed target budget
Frequency support shift	8	Target frequencies shifted beyond source support
Coordinate permutation	8	Ordered values are permuted before implementation
Irregular grid	8	Nonuniform declared nodes
Piecewise smooth	12	Blockwise basis rather than cosine basis
Sparse high frequency	16	Few active high-frequency coordinates
Misspecified target	24	Target center independent of source centers

Table 3 Primary randomized profile results. Each row contains 480 paired task–resolution cells: 160 independently seeded latent tasks crossed with three grid resolutions. Standardized DCT maximin and the source-greedy portfolio are post-review supplementary comparisons on the original tasks.

Initial design	Tasks	Feasible (%)	Certified (%)	False cert.	Penalized loss
Unstandardized DCT maximin	480	37.5	10.6	3	0.746
Natural blockwise	480	83.8	19.4	0	0.278
Random low frequency	480	50.8	13.5	0	0.610
Raw Sobol	480	80.2	30.0	2	0.337
Source-scored atlas	480	91.7	47.3	0	0.163
Source-greedy portfolio	480	87.1	37.1	3	0.178
Standardized DCT maximin	480	81.0	31.5	0	0.274

Table 4 Prospective confirmation on 240 new latent tasks across three fixed source families and public libraries, at $d = 1000$ and $n_0 = N = 10$. Coverage counts designs with a truly feasible policy; true certificates exclude the one false certificate recorded by each design. Medoid + structural is standardized generic DCT maximin.

Design	Coverage	True certificates	False	Loss
Medoid + augmented	137/240	24/240	1	0.518
Source-scored atlas	155/240	56/240	1	0.414
Source-best + raw L2	148/240	46/240	1	0.442
Source-best + structural	155/240	54/240	1	0.414
Source-greedy portfolio	212/240	49/240	1	0.156
Medoid + structural	138/240	24/240	1	0.515

Table 5 Rates aggregated over three grid resolutions within each regime. The comparator is the best source-free design in that row.

Regime	Source feasible	Source certified	Best source-free certified	Difference
Aligned low frequency	100.0	56.7	Raw Sobol: 91.7	-35.0
Growing rank	100.0	40.0	Random low frequency: 20.0	+20.0
Frequency shift	95.0	60.0	Standardized DCT maximin: 65.0	-5.0
Permutation	100.0	90.0	Random low frequency: 50.0	+40.0
Irregular grid	100.0	15.0	Raw Sobol: 38.3	-23.3
Piecewise smooth	100.0	95.0	Standardized DCT maximin: 95.0	+0.0
Sparse high frequency	63.3	21.7	Natural blockwise: 25.0	-3.3
Misspecified target	75.0	0.0	Raw Sobol: 1.7	-1.7

Table 6 Controls for the information available at $d = 1000$.

Schema	Source weighting	Tasks	Feasible (%)	Certified (%)	Loss
declared	domain blind	160	95.0	46.2	0.143
declared	conditioned	160	85.0	44.4	0.258
schema blind	domain blind	160	92.5	35.0	0.180
schema blind	conditioned	160	82.5	33.1	0.296

Table 7 Equal preverification cost at $d = 1000$: 384 source plus 10 target calls versus 394 target-only calls. Verification remains additional and is included in the final column.

Method	Source	Target	Certified (%)	False	Algo. fail.	Mean all-in
Unstandardized DCT maximin	0	394	53.8	0	0	564.5
Natural blockwise	0	394	53.1	0	0	564.5
Random low frequency	0	394	45.6	0	0	584.0
Raw Sobol	0	394	53.8	0	0	564.0
Source-scored atlas	384	10	46.2	0	0	562.0
Target-only functional SCBO	0	394	58.1	0	1	549.1

Table 8 Outcome-adjusted simulator cost. Calls per success equal mean amortized all-in calls divided by the observed true-feasible certification rate. Break-even compares source scoring with each alternative, charging both source-based designs for their archive.

Matrix	Design	Certified (%)	Calls/success ($M = 1$)	Calls/success ($M = 20$)	Break-even M
$N = 10$ primary	Unstandardized DCT maximin	10.6	2192.9	2192.9	1
$N = 10$ primary	Natural blockwise	19.4	1143.2	1143.2	2
$N = 10$ primary	Random low frequency	13.5	1716.9	1716.9	1
$N = 10$ primary	Raw Sobol	30.0	709.4	709.4	3
$N = 10$ primary	Source-scored atlas	47.3	1182.7	411.3	—
$N = 10$ primary	Source-greedy portfolio	37.1	1596.4	612.7	1
$N = 10$ primary	Standardized DCT maximin	31.5	639.5	639.5	4
Equal preverification	Unstandardized DCT maximin	53.8	1050.2	1050.2	2
Equal preverification	Natural blockwise	53.1	1062.6	1062.6	2
Equal preverification	Random low frequency	45.6	1280.0	1280.0	1
Equal preverification	Raw Sobol	53.8	1049.3	1049.3	2
Equal preverification	Source-scored atlas	46.2	1215.1	426.4	—

Table 9 Corrected Energy V5 region-held-out results over 18 markets and five seeds per market. All policies receive a common 50% initial SOC; every design uses the same finite service objective and empirical-window verifier. Objective is the median of market medians among each design's certified safe runs; designs can therefore contribute different subsets of markets.

Design/backend	Certified	False	Median objective	Mean verify
Source-scored atlas	57/90	0	0.0274	153.8
Standardized DCT maximin	49/90	0	0.0346	167.1
Source-greedy portfolio	59/90	0	0.0273	160.0
Unstandardized DCT maximin	54/90	0	0.0406	168.0
Random low frequency	56/90	0	0.0301	163.6
Natural constant grid	52/90	0	0.0253	176.0
Raw Sobol	43/90	0	0.0701	180.4
Target-only functional SCBO	51/90	0	0.0391	177.8